IMMUNITY PROTOCOL

AI CODING WORKSHOP: DEBUGGING & ADVANCED POLISH

(YOU)us

MACQUARIE University
SYDNEY · AUSTRALIA

Acknowledgements

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BIOPLATFORMS AUSTRALIA

NCRIS National Research Infrastructure for Australia
An Australian Government Initiative

australian genome foundry

APAF
australian proteome analysis facility

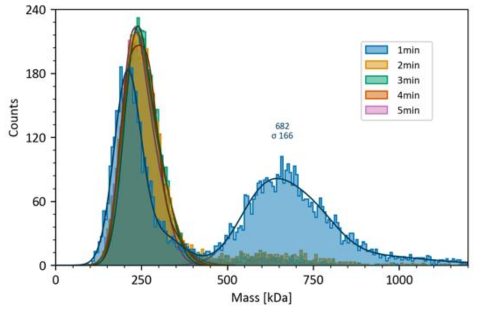
Australian Proteome Analysis Facility

The Australian Proteome Analysis Facility (APAF) is part of the Macquarie Analytical and Fabrication Facility (MAFF), providing equipment and support for training, teaching, and research.

APAF supports research across fields such as chemical and biological sciences, biomedicine and health, food and agriculture, and microbiology.

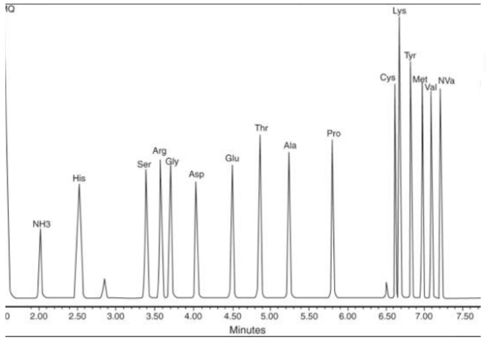
Mass spectrometry

- Bottom up and top down
- Protein coverage, identification, quantification
- PTM analysis
- Intact protein characterisation



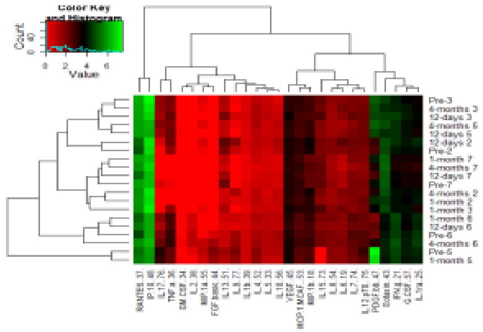
Protein Interactions

- Mass photometry in solution analysis
- Mass spectrometry based interaction identification
- Co-IP mass spectrometry
- Array technology, immunoassays



Protein Analysis

- Amino acid analysis
- Gel electrophoresis
- High performance liquid chromatography (HPLC)

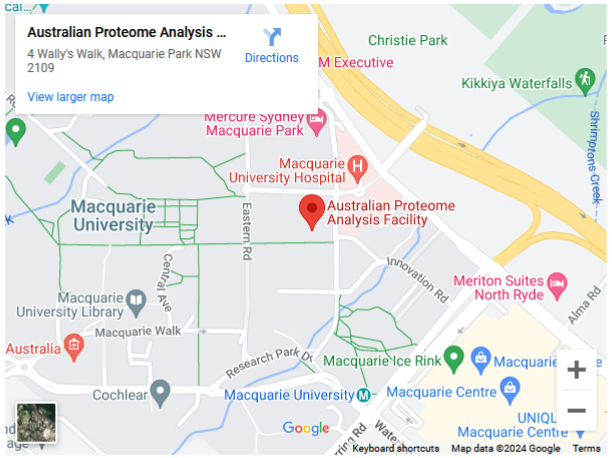


Bioinformatics

- Tool development
- Data analysis and collaborations
- Data mining and classification
- Biomarker development pathway analysis

Sydney

Australian Proteome Analysis Facility
Level 4, 4 Wally's Walk
Macquarie University NSW 2109
Phone: +61 2 9850 6201
Email: info.apaf@mq.edu.au



Bioinformatics Consult at APAF

Contact Ignatius Pang

Ignatius.pang@mq.edu.au

Every Tuesday 1 – 2 pm, by appointment only

PRE-WORKSHOP SETUP

MACQUARIE University SYDNEY·AUSTRALIA

1. Create a Google Account
2. Install Google Chrome Browser
3. Install Antigravity IDE
4. Download and Unzip the AI Workshop Repository
5. Disable Telemetry in Antigravity IDE
6. Install & Configure IDE Plugins

Remember to record the workshop

CODES DOWNLOAD

The text prompts and starting scripts for the workshop can be found in https://github.com/APAF-bioinformatics/AI_Workshop

Workshop Overview & Objective

Goal: Recreate “Immunity Protocol” — a cellular-themed defensive space shooter.

Technologies: HTML5 Canvas, Web Audio API, Vanilla CSS, modern JavaScript.

Focus Areas:

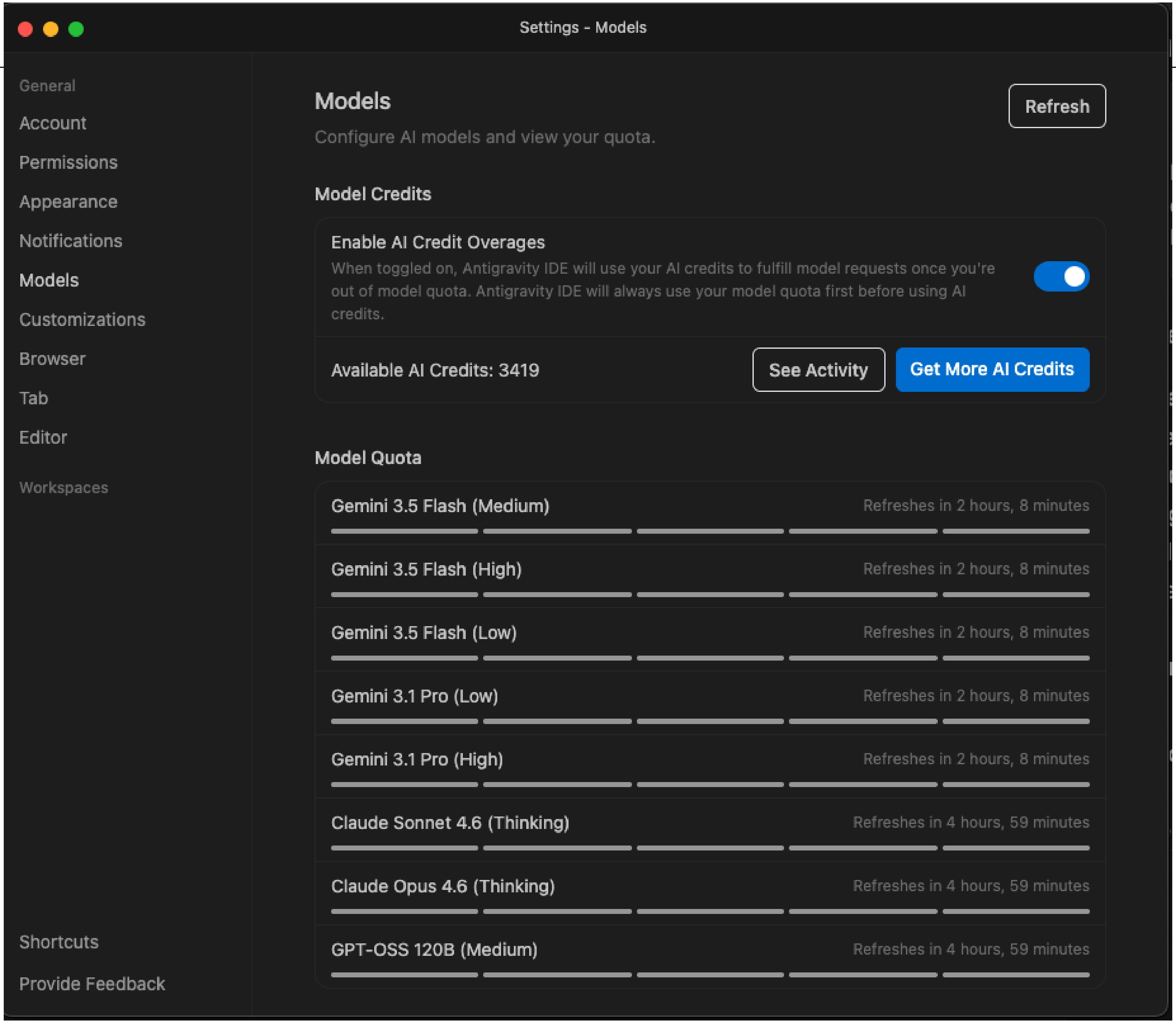
1. Core physics (fluid movement through viscous medium).
2. Procedural audio synthesis (no external library, clean synthesis).
3. FSM state design (boss behaviors, variant waves).
4. Design aesthetics (microscope grids, light particle systems).

Agenda & Setup (On-the-Day)

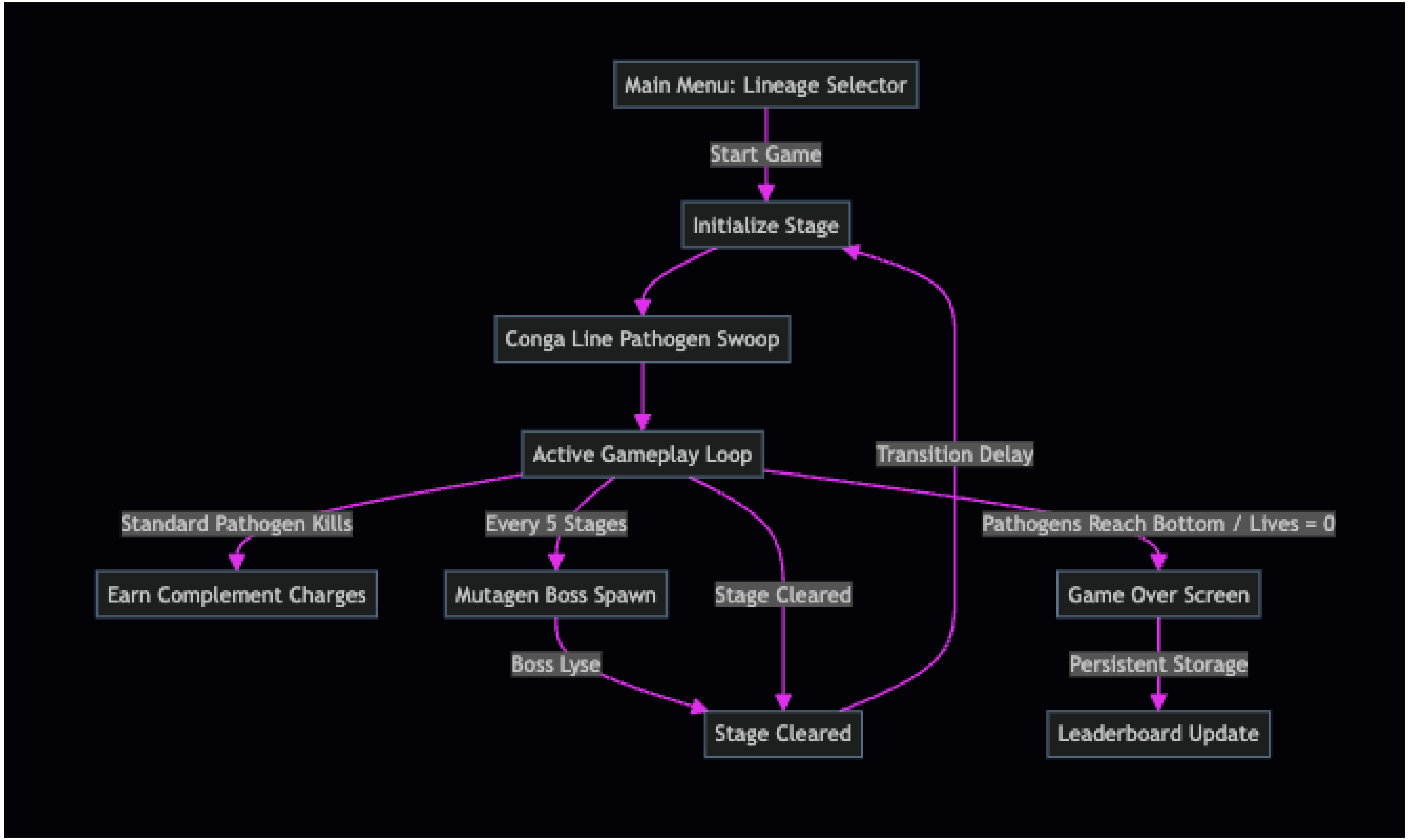
1. **Repository Setup:** Clone the repository and unzip.
2. Due to limited AI tokens, we had already used all the prompts in data/game_prompts.md to create a starter version of the game
3. **Starter File Placement:**
   - Copy `data/index_to_debug.html` to the root folder.
   - Rename it to `index.html`.
4. **Core Task:** Resolve 5 critical bugs in `index.html` (Tasks A–E).
5. **Data Privacy:** Go through *.aiignore* and *.gitignore* files
6. **Advanced Polish:** Implement optional enhancements (Prompts 16–24) using the custom `impeccable-design` system rules.

How to view AI tokens usage?

- If on Google Pro-plan, Gemini tokens replenishes fully every 5 hours
- If on NO plan, replenishes fully every 5 days
- Check it at least after every second task



Architecture Flowchart



CORE DEBUGGING TASKS

Reporting error on messaging channel

ALSO DEMONSTRATE HOW TO USE THE ANTIGRAVITY PROMPT

- **Show how to change the model**
- **How to send a command**
- **How to attach files**
- **How change model**
- **Systematically go over all the buttons around the prompt**
- **How to submit and stop**

Error reporting template

- ***Task:*** [e.g., Task C - Barrier Shield]
- ***Error/Symptom:*** [e.g., Bullets pass through early / console error]
- ***Workspace State:*** [e.g., Using Antigravity IDE, AI checking browser console for errors]
- Put the above error reporting message into the Antigravity now

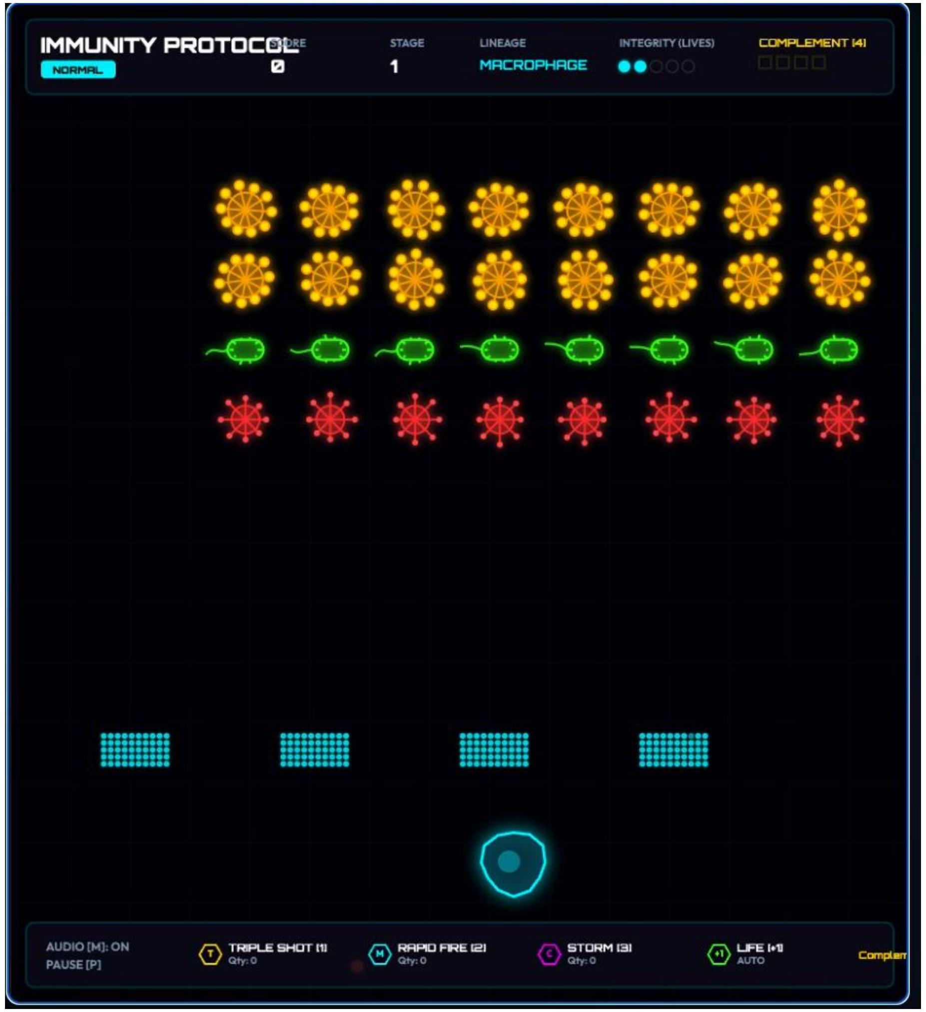
Core Debugging: Tasks A & B

🚀 Task A: Jerky Player Movement

Problem: Horizontal cell movement is jerky and changes coordinates erratically.

Goal: Replace unstable position updates with linear interpolation (lerp) or damping:

```
targetDir += (moveDir - targetDir) * dt * 15.0
```



Core Debugging: Tasks A & B

📊 Task B: Complement UI Alignment

Problem: Complement HUD tracker text overflows past the right side of the screen boundary.

Goal: Align margins and positioning coordinates inside the HUD render loop.

We need to fix the text as it runs outside the frame

Core Debugging: Task C (Barrier Collision)

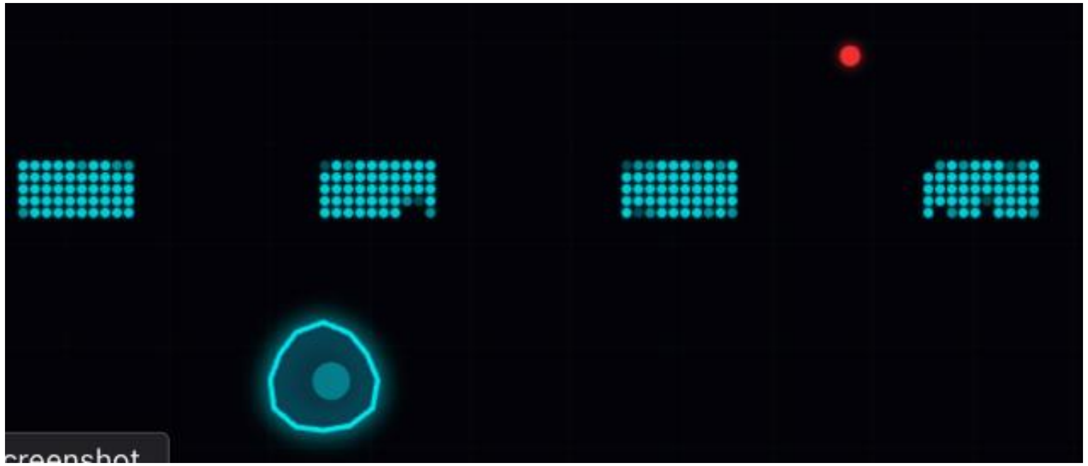
Layer-by-Layer Damage

Problem: Damage applies out of order (starting at row 2 instead of the row closest to the projectile).

Rules:

- Player bullets damage bottom-most block in a column.
- Enemy bullets damage top-most block in a column.
- Bullets cannot pass through until the column is empty.

Check Quota: Now is a good time to check your quota usage.



Pathogen Grid & Barriers

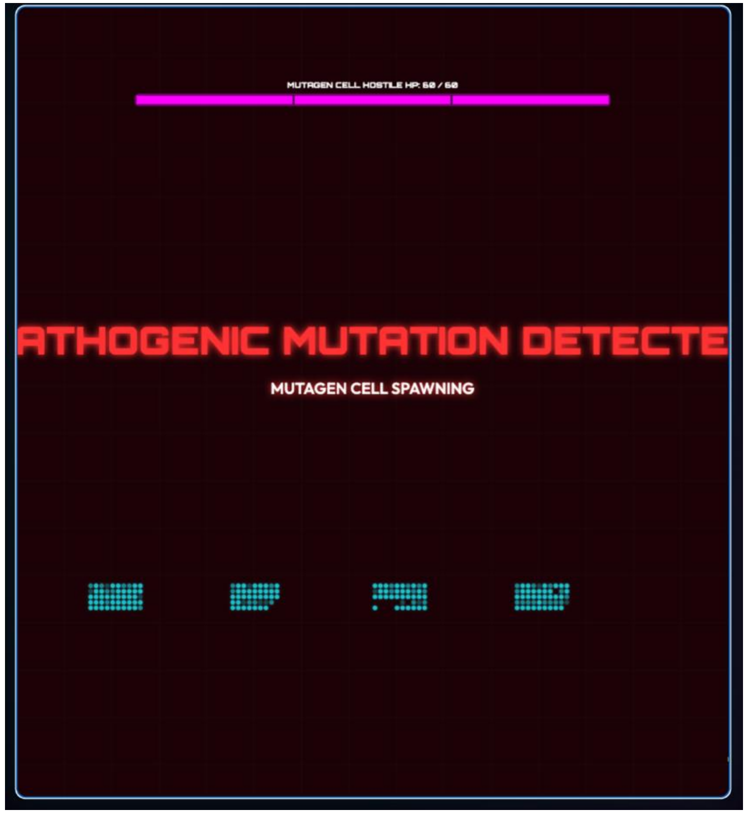
Core Debugging: Task D (Boss Spawn Freeze)

👾 Boss FSM Rescue

Problem: Spawning the stage 5 boss causes the game loop to freeze.

Shortcut for Testing: Bind key `8` to skip straight to the next boss level (Stage 5, 10, 15, etc.).

Note: Ensure shortcut listeners are removed after testing.

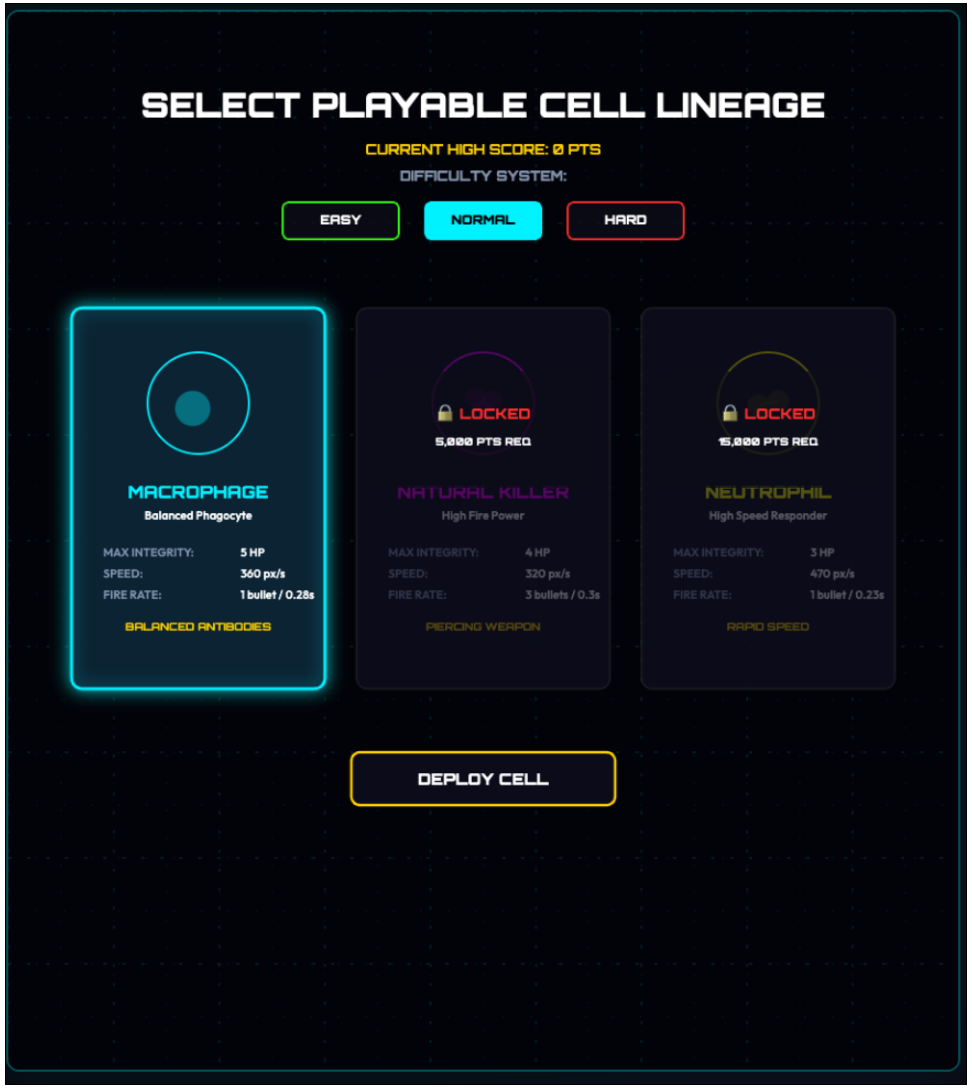


Mutagen Cell Spawn Warning

Core Debugging: Task E (Lineage Unlock Logic)

Goal: Lock the **Natural Killer** (NK) cell and **Neutrophil** cell lineages initially.

Unlock Threshold: Enable lineage selection only when the persistent player high score surpasses the unlock threshold (15,000 points).



Testing Notes: Managing Console & High Scores

Open DevTools Console (`F12` or `Cmd+Option+I` on Mac) to test locks:

Reset High Score Only:

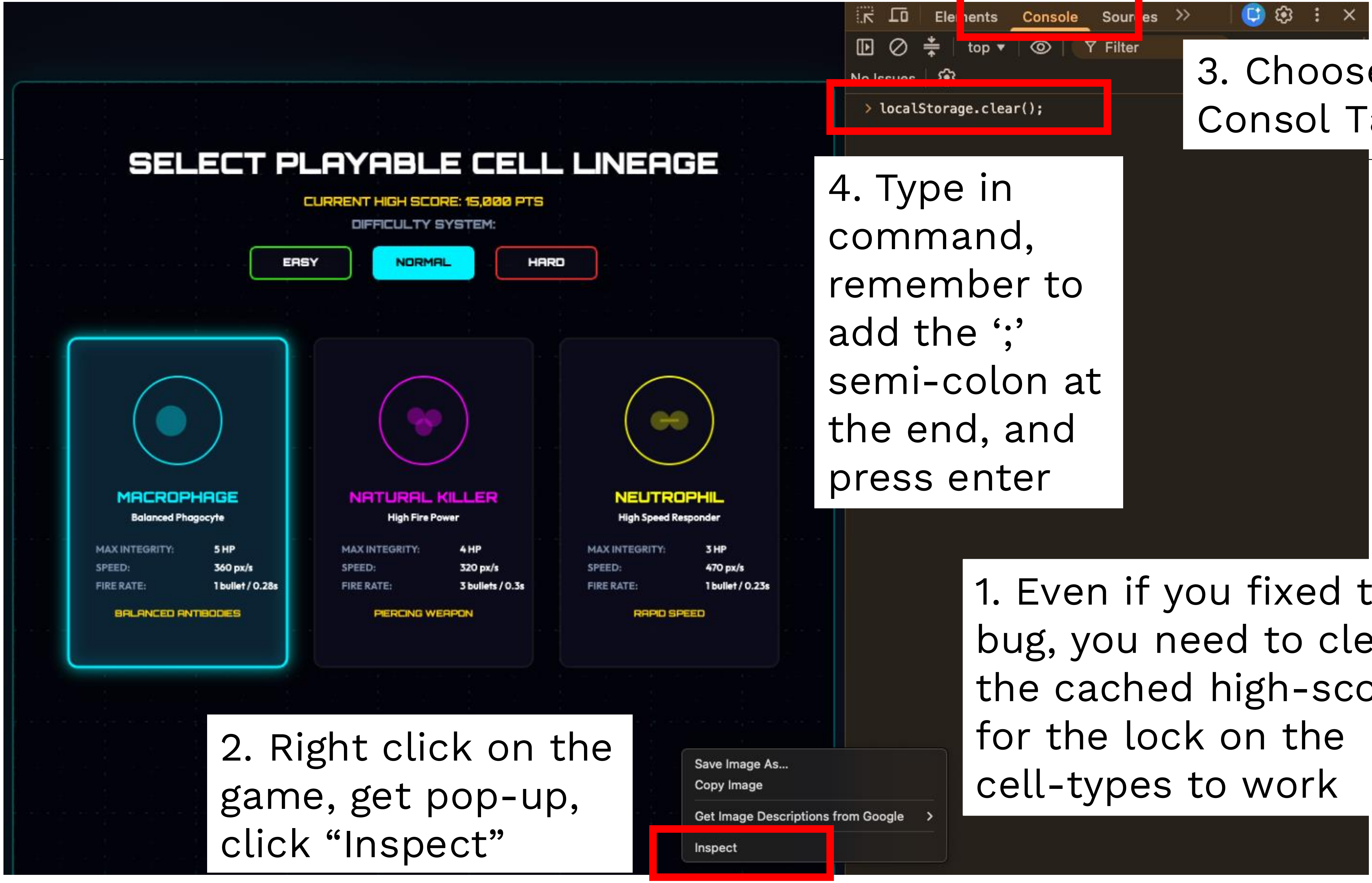
```
localStorage.removeItem('immunity_highscore');
```

Reset Everything (High Score & Leaderboard):

```
localStorage.clear();
```

Unlock All Cells Instantly (Set Score to 15,000):

```
localStorage.setItem('immunity_highscore',
'15000');
```



1. Even if you fixed the bug, you need to clear the cached high-score for the lock on the cell-types to work

2. Right click on the game, get pop-up, click "Inspect"

3. Choose the Consol Tab

4. Type in command, remember to add the ';' semi-colon at the end, and press enter

The First Session Handover

- Please load session handover skill and save session handover file in the session_handover directory with the suffix _YYYYMMDD_HHMM.md

AI Coding Course Workshop Feedback

https://forms.cloud.microsoft/r/UTdY3rB6JD

ADVANCED POLISH & EXTENSION PROMPTS

Optional Advanced Polish: Design System (Task F)

Ensure your additions follow the **Impeccable Design** system rules:

Motion Design: Use custom cubic-bezier deceleration curves (`cubic-bezier(0.22, 1, 0.36, 1)`) for drift/UI animations instead of linear steps.

Color Uniformity: Define visual asset colors in the perceptually uniform OKLCH space. Use tinted brand neutrals instead of flat grays.

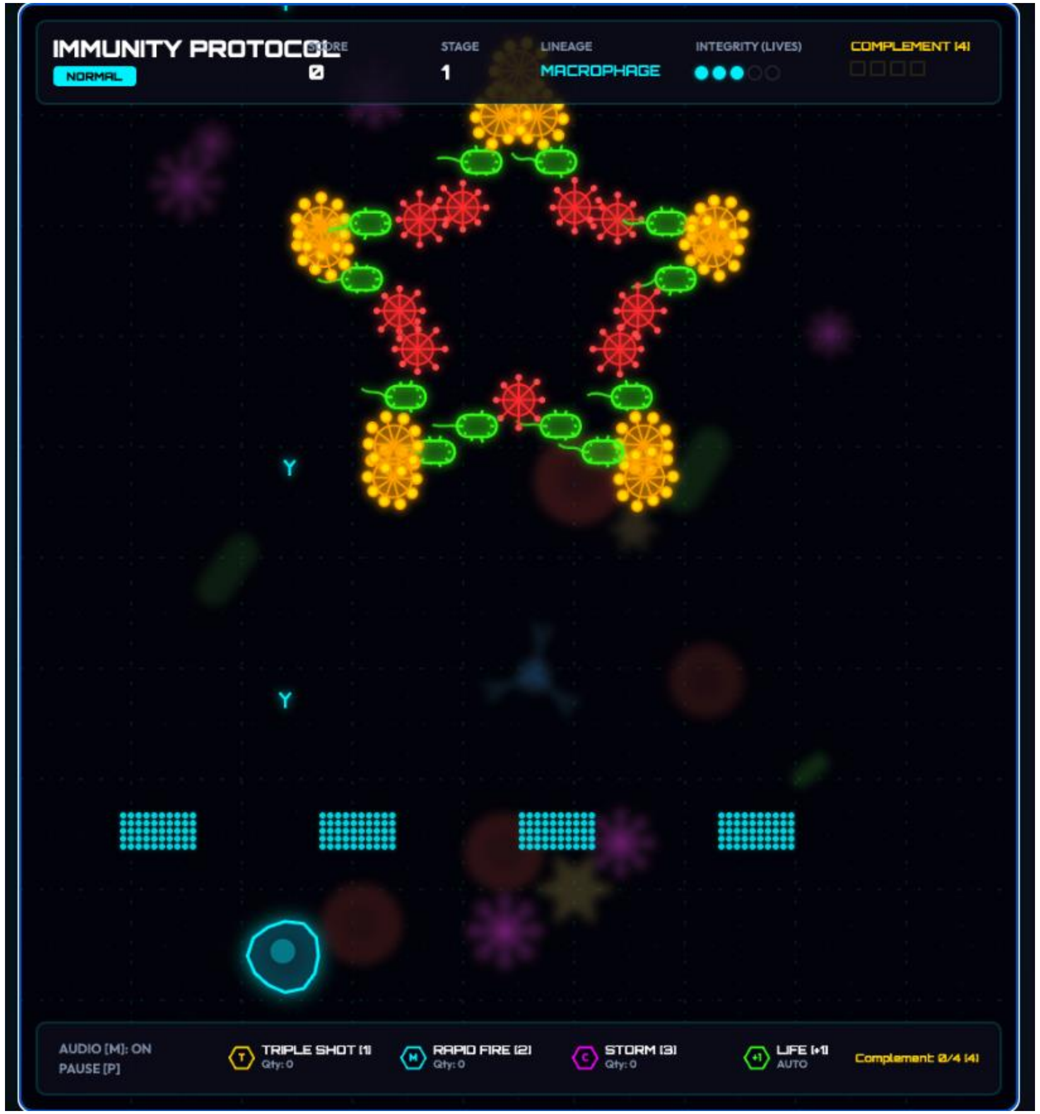
Design Assets Map:

- Motion Guidelines: `.gemini/skills/impeccable-design/reference/animate.md`
- Color Rules: `.gemini/skills/impeccable-design/reference/colorize.md`

Background and Microscope Grids (Task G)

Task G (Microscope Grid & Specimen Drift):

- Double-layered dotted reticle grids (`[3, 15]` and `[2, 24]` dash spacing).
- Out-of-focus background speculation elements (platelets, cells) drifting slowly with canvas filter blur: `ctx.filter = 'blur(4px)'`

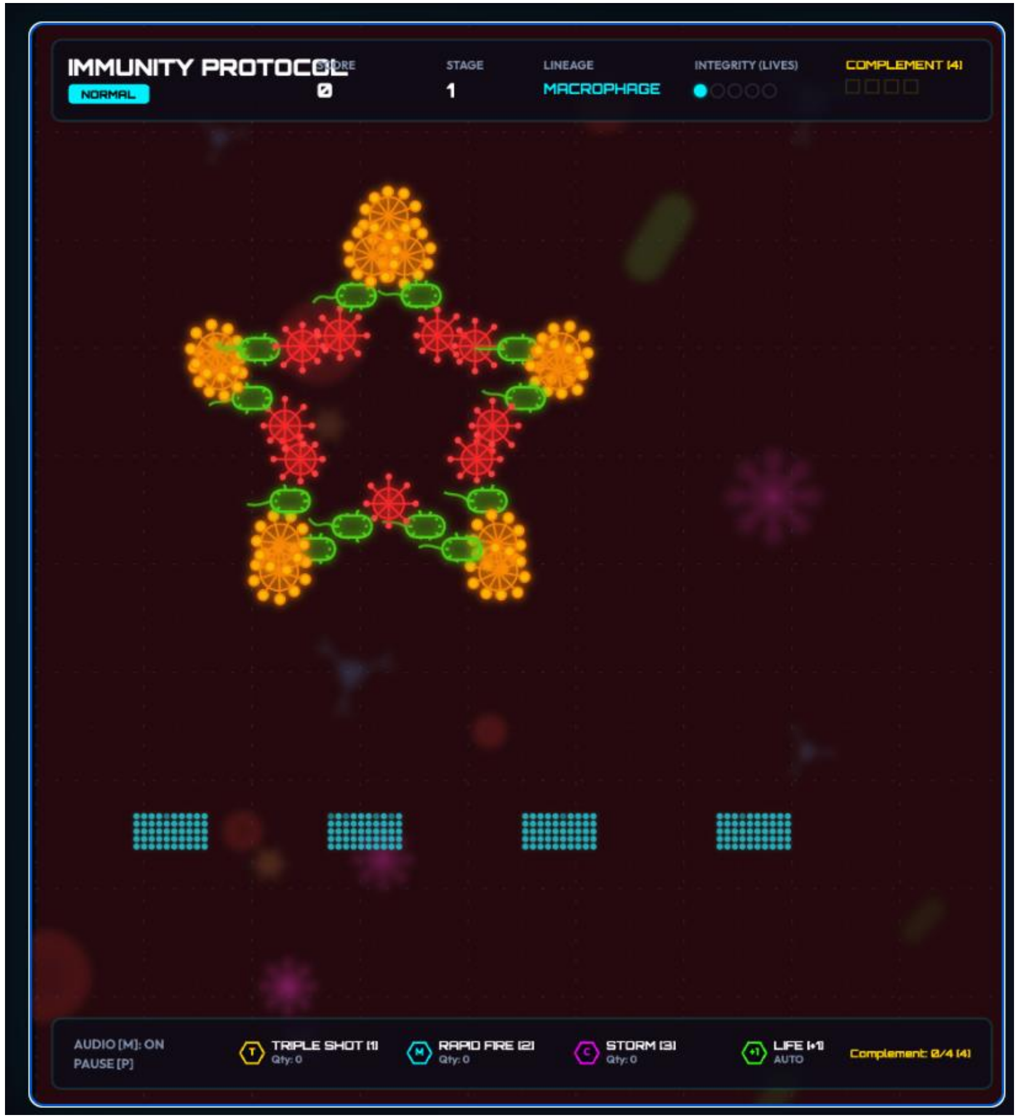


Damage Warnings & Bullets Paths (Tasks H – I)

Task H (Screen Shake & Episafe Blink):

- Screen shake matrix offsets triggered on lysis and damage events.
- Episafe red life-loss screen blink (`sin(ratio * π) * 0.2` opacity).

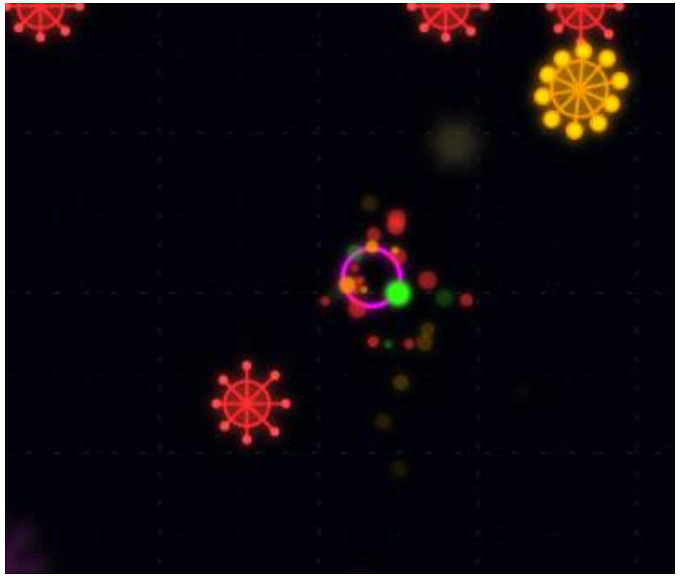
Task I (Fluid Bullets): Sinusoidal sideways drift added to update cycles to simulate trajectory movement in thick cytoplasm.



Particle Emitters & Boss Extensions (Task J - K)

Task J (Lightweight Particle Engine):

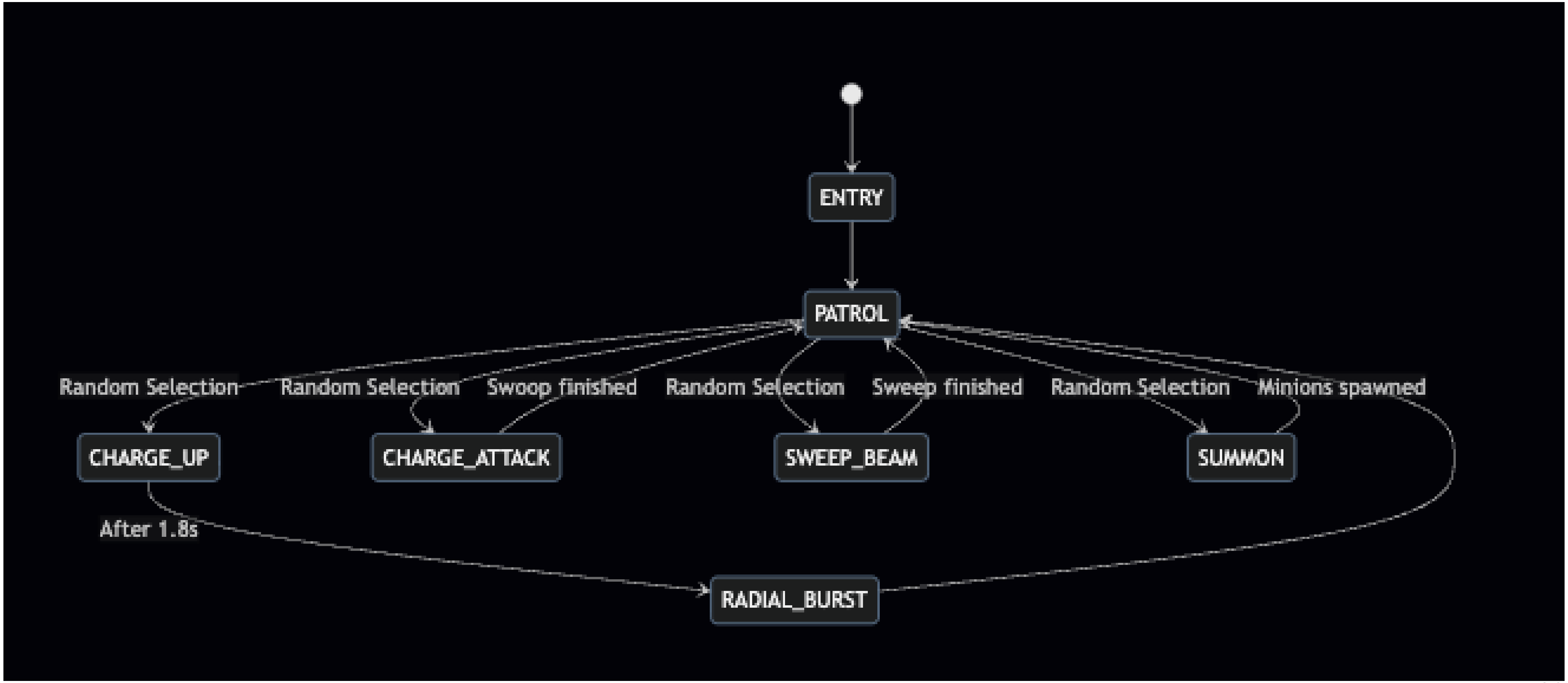
- Fading particle bursts triggered during pathogen lysis (explosions) and player cell hits.
- Must control angle spreads, velocity decay, and canvas alpha decays cleanly.



Task K (Boss Finite-State Machine Extensions):

- `SWEEP_BEAM`: Side-to-side sweeping while firing 5-bullet downward spreads (`-90°` to `+90°`).
- `SUMMON`: Spawn helper Virus minions (max 8 on screen).
- *Bug Prevention:* Clear all remaining minion objects from memory immediately when the boss is lysed.

Boss State Transition Map (With Sweep & Summon)



Conga Line Swoop & Shielded Boss (Task L - M)

Task L (Conga Line Entry Choreography):

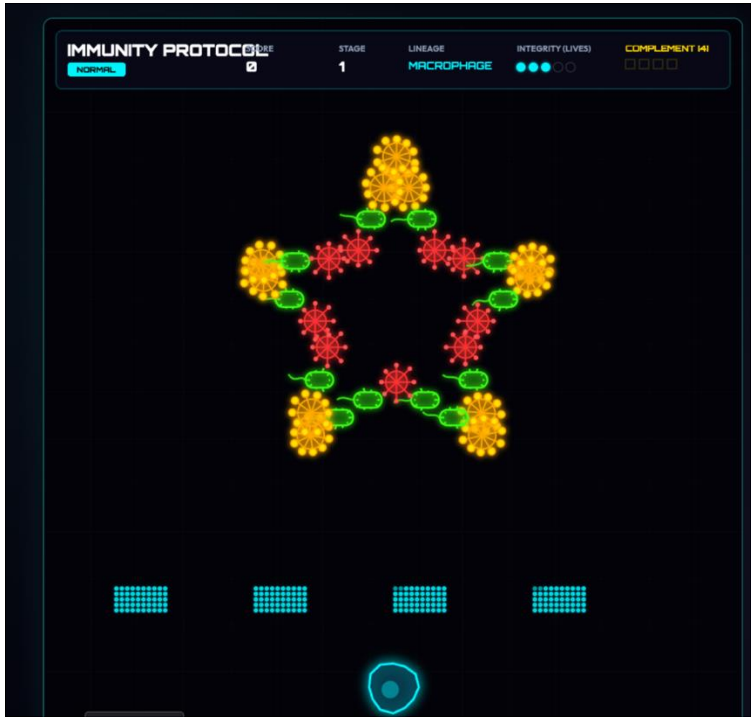
- Pathogens swoop onto the screen one by one from top-left/top-right.
- Perform a `360-degree loop` at the screen center before drifting to their home grid slots.

Task M (Bio-Carrier Overlord):

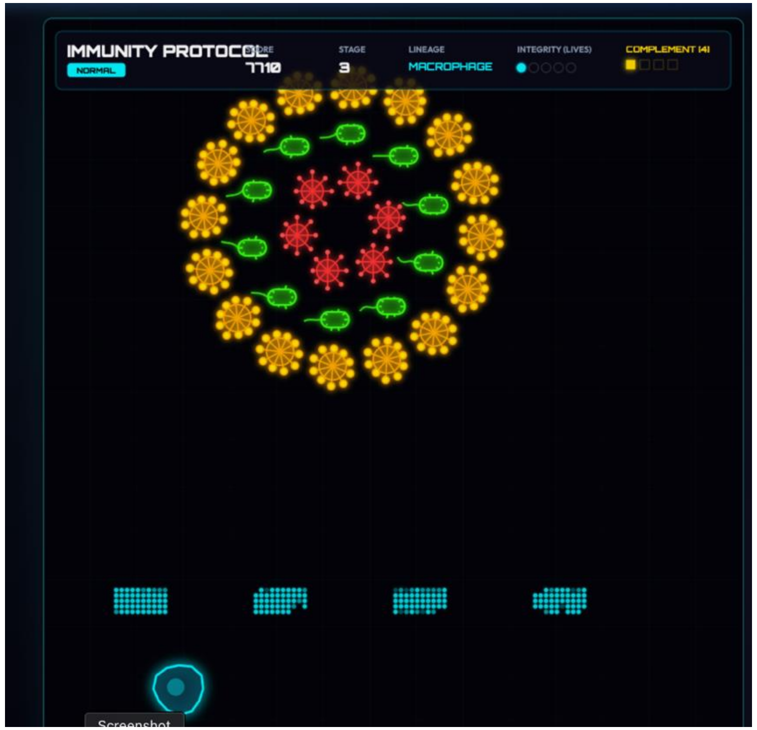
- **Phase 1:** Protected by 4 orbiting generator nodes (10 HP each). Boss is completely immune to damage while nodes live.
- **Phase 2:** Nodes collapse, triggering magenta color shift, +40% speed boost, and homing projectile target locking.

Advanced Formations (Task N)

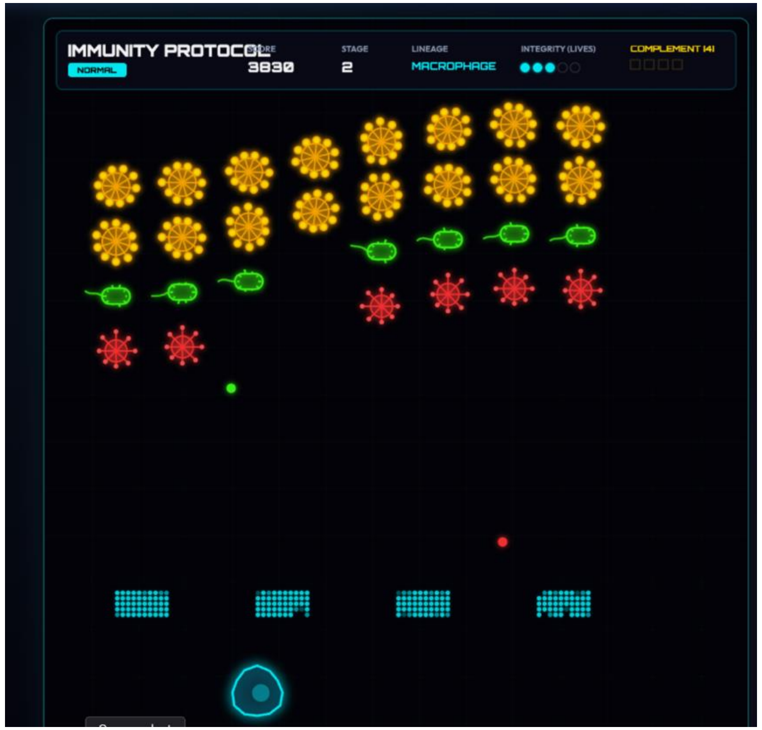
Pulsating Star



Spinning Circle

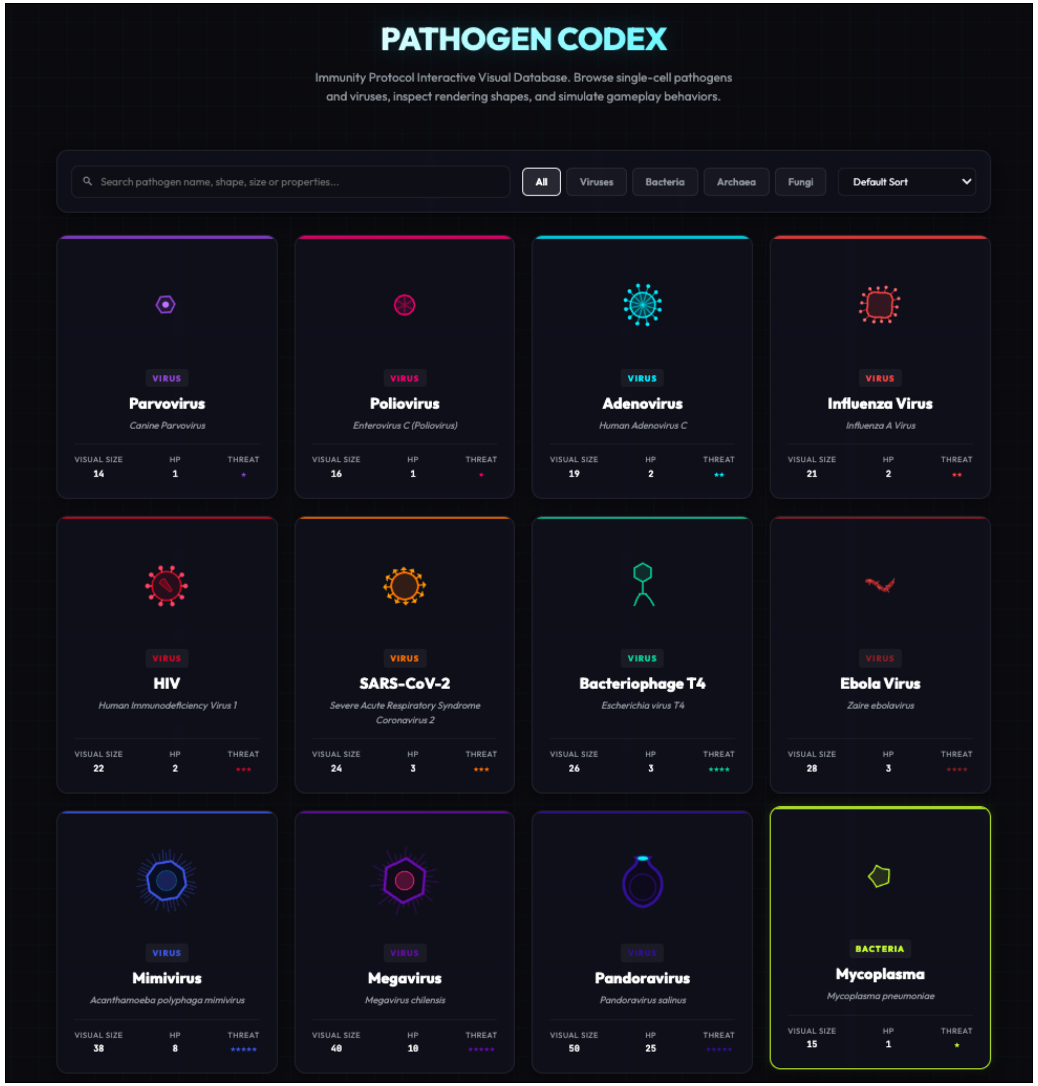


Undulating Wave Shape



Add more varieties of pathogens (Task 0)

- We have a library of pathogens in ‘*pathogen_showcase.html*’ and their shapes pre-prepared for us to conserve tokens
- Use these to populate the different levels in the game



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